

ArUco markers pose estimation in UAV landing aid system

Adam Marut, Konrad Wojtowicz, Krzysztof Falkowski

Faculty of Mechatronics and Aerospace,

Military University of Technology

Warsaw, Poland

adam.marut@wat.edu.pl, konrad.wojtowicz@wat.edu.pl, krzysztof.falkowski@wat.edu.pl

Abstract— Recently, UAV usage spread to various areas of life. In many cases, UAVs carry expensive equipment onboard. Most of the aircraft damage occurs during the landing due to hard landings. To decrease the number of such situations landing aid systems are being developed. Since most of those systems need special airfield infrastructure, in this project a simple and low-cost system is proposed. This paper provides an overview of the UAV landing aid systems and a concept of the computer vision-based landing aid system. The proposed solution uses ArUco markers as reference points during the landing. By using marker pose estimation algorithm it is possible to determine the height of the flight during the touchdown with high accuracy. The fusion of the data acquired from the inertial sensors generates precise information about the path of the approach of the UAV. In addition, markers placed across the airstrip may be used to navigate during taxing. The developed solution has been tested for accuracy of position determination and for an influence of marker size on accuracy and measurement divergence.

Keywords—landing aid, ArUco markers, UAV, machine vision

I. INTRODUCTION

Nowadays, in industry, the usage of UAVs has highly increased. Variety of types of drones gives the opportunity to use them in many different tasks. The most common applications include small package transport, terrain mapping for the geodetic needs, terrain observation or just recreational flights for enthusiasts. In every case, the flight can be divided into three main phases:

- Take off;
- Mission execution;
- Landing;

For long-range flights, there is currently a lot of emphasis on making the flight autonomous or at least semiautonomous. These autonomy features rely heavily on implemented navigation systems. As take-off and mission itself are considered rather safe, the landing turns out to be the most dangerous stage of the flight. Most of the damage of UAV occurs during this maneuver. The main issue is that the GPS is not a reliable way of determining the height of flight and its accuracy depends on the surroundings, which can disturb the signal. On the other hand, ultrasonic height sensors are not credible on grass airfield due to characteristics of the signal.

A. Overview of the landing aid systems

The most popular landing aid system is ILS (Instrument Landing System) developed in the first half of the 20th century. The system generates one glide slope by using radio signals of two frequencies (150 Hz and 90 Hz). The receiver on the aircraft detects the middle of the beams which represents the optimal glide slope. The biggest advantage of this system is the ease of use and long operating range. On the other hand, it needs special airfield infrastructure and it proved to be susceptible to electromagnetic interference in its surroundings. However, there is a number of restrictions in use with UAVs. This system is only implemented on large airports, which excludes small UAVs operating on grass airfields. Moreover, this system gives information on approach only. Depending on ILS category pilot continues landing manually on decision height which is 60 m for I category and 15 m for IIIB category. It means that the pilot does not receive any help from the ILS system during a touchdown. In 1970s MLS (Microwave Landing System) was presented. It had a lot of advantages over the ILS – multiple glide slope, more channels etc. However, this system development failed due to the rise of satellite navigation systems. Nowadays, the most popular navigation systems are GNSS, but these systems are not accurate enough to be the only aid during landing.

B. UAV landing challenges

UAV flights can be divided into two main categories: within pilot sight (VLOS) and beyond visual line of sight (BVLOS) by using FPV (First Person View) and automatic navigational equipment. This paper covers BVLOS flights challenges. The major challenges during the landing phase of flight are:

- The difficulty in unambiguous determination of the position in space basing on the view from the FPV camera;
- Without any landing aid information, it is hard to determine the precise height over the airstrip;
- For the airfield without proper infrastructure it is hard to determine the position of the aircraft on the runway or on the taxiway;

In this matter in research conducted by Nanjing University of Aeronautics and Astronautics automatic landing of UAV is based on cooperative target which is made of infrared emitting T-shaped sign [1]. The proposed method is suitable only for hovering UAVs rather than for fixed-wing drones which encounter difficulties as above. This system can be used for locating and indicating the direction of the runway from the air. However, it provides no information during final approach and landing phases. A similar problem was discussed in the research on the vision-based system [2], but it also did not apply to fixed-wing UAV. Another approach to the pose estimation is inertial sensor employed during the landing. To achieve precise position estimation with this method, high-quality and expensive inertial sensors are required. Nevertheless, in research [6] an error of the determined location was higher than 10 meters. Such an error is way too high for an automatic landing.

This paper shows the concept of the computer vision-based landing and navigation aid system for fixed-wing UAVs. The system contains graphical markers placed along both sides of the runway. The second section of the paper describes the chosen graphical markers, pose estimation algorithm and positioning of the markers on the airfield. Section III covers the initial tests of the pose estimation when using a single marker. The article is summarized with conclusions and future work outlook.

II. DESIGN CONCEPT

The main task of the proposed solution is to help the pilot of a UAV during the final approach. Moreover, after the touchdown, the system will give information about the position on the runway or taxiway. The system is based on computer vision algorithms implemented in ArUco module from the OpenCV libraries.

A. Markers detection and pose estimation algorithms

ArUco markers are designed to provide quickly a 3D position of the camera towards the single marker. The method is highly resistant to false detection. [4] Detection algorithm was optimized to receive a low false negative rate of detection by Hamming coding algorithm. The detection process is divided into stages as follows:

- Receive an image from the camera (Fig.1a);
- Change image to grayscale;
- Contour extraction (Fig.1b, 1c); and filtering to get only marker candidates from the image – rectangles (Fig.1d);
- Perspective removal (Fig.1e);
- Extracting the binary code on the marker (Fig.1f) and comparing to the markers dictionary;
- Generating id of the marker and the rotation and translation vectors (to determine the position of the marker);

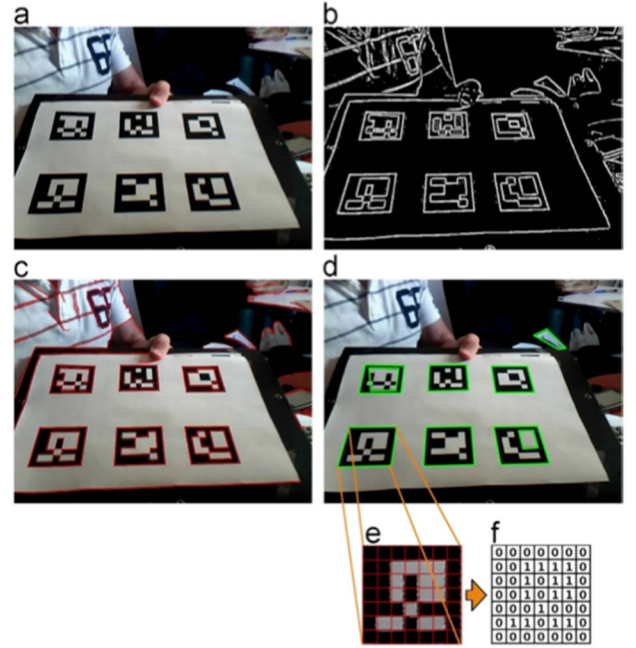


Figure.1 Detecting process conducted on the image [4]

Inner matrix of the marker is coded with the Hamming algorithm. It provides a single bit error detection and correction in every binary line. Markers IDs are defined in the dictionary, which can be predefined in ArUco module or manually created by the user. After the id detection, the solvePnP function is employed for every corner of the marker. At this stage, it is essential to provide the detailed camera optics parameters and distortion matrixes. Those values are collected in the camera calibration process. To receive reliable results the optics of the camera has to be static. After solving the following equation the 3D position of the point is determined from the 2D image.

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{pmatrix} f_x & \gamma & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X_G \\ Y_G \\ Z_G \\ 1 \end{bmatrix} \quad (1)$$

Where the first vector describes the position of the point on the image (u, v), two following brackets stay for the optics parameters and the X_G, Y_G, Z_G describe the point in space based on a camera reference system. After placing the plane through the four points the algorithm determines vectors of rotation and translation between the camera plane and the marker plane. Due to the speed of the marker recognition algorithm and the accuracy of determining their position, the method can be used in the proposed landing aid system.

B. ArUco state of art.

ArUco markers were originally developed for AR applications. Because of its features, markers are used in cases where pose estimation is needed. In most cases, ArUco markers are used in the indoor navigation systems or in an environment where GPS signal does not provide the

desired accuracy. In most researches, there are used for quadrotors or helicopter [2] [3] [7]. They are also used in other applications such as limb positioning during medical rehabilitation [4].

C. Airstrip infrastructure

Proposed landing aid system will consist of portable signs which will be placed at equal distances along the runway. Every sign will have its individual ArUco marker. Information about the markers ID and their position will be added to the UAV navigation system. Information about the position of the aircraft will be calculated by using translation vectors from ArUco pose estimation algorithms. In the result of the calculations, the determination of the glideslope and altitude over the runway is possible. Precise information about altitude is crucial during the touchdown and can decrease hard landing occurrence. This feature of the system can increase the safety of FPV flights. In addition, the pilot operating on UAV will have precise information about the position on the runway. This value can be used in the situation when the aircraft is rolling too fast on the runway (estimated braking distance is longer than remaining runway length) and has to go-around to prevent damage of the UAV. In the situation of the correct landing, the pilot can adjust the braking power.

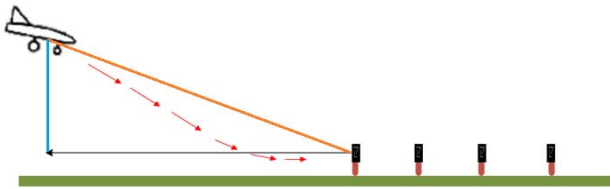


Figure 2. The concept of the approach to the airfield

As presented in figure 2, during the UAV final approach the image from the camera mounted on the nose of the aircraft will be analyzed against ArUco detection. The pose of the UAV will be calculated based on the vectors to visible markers. With this information, it will be possible to estimate the height over the marker. Subsequently, the relative height over the runway will be calculated as follows:

$$h = h_{dif} + h_{aruco} \quad (2)$$

Where:

h – relative height over the runway;

h_{dif} – height of UAV over the middle point of the ArUco marker;

h_{aruco} – height of the marker sign (distance from the ground to the middle point of the ArUco marker);

After gaining the desired height, horizontal vectors are considered to estimate the position of the runway axis. The fusion of the position data towards consecutive markers gives the complete information required for the landing. The

conception of infrastructure architecture is presented in figure 3.

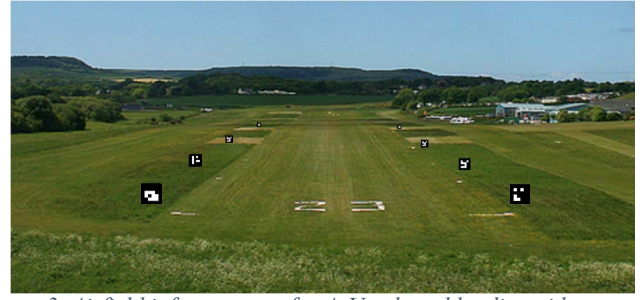


Figure 3. Airfield infrastructure for ArUco based landing aid system

As shown in the figure above at least four markers are visible in the image from the camera during the landing. This gives the higher accuracy of the pose estimation of the UAV.

D. Hardware implementation

The concept of this system has to be verified in various scenarios. As the system is dedicated to UAV, it has to be developed as an embedded system. For the test, the ODROID-XU4 (figure 4) was chosen, as it is one of the most powerful microcomputers on the market. To provide long-range detecting, which is crucial during the landing, the camera with the interchangeable lens was used for the test.

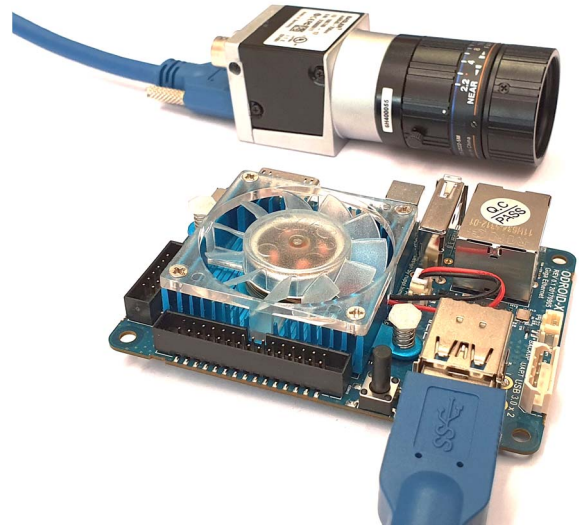


Figure 4. Odroid XU4 with a Basler ACE camera connected via USB 3.0

The tested camera has also a high frame rate to satisfy the dynamic requirements of the system. It is crucial to have the optical system which cannot be easily disrupted by the sun or other light sources, the CMOS image sensor of the camera is resistant to local overexposure. The computing unit provides the 10 Hz refresh rate of the image. Conducted verification tests cover the accuracy of the pose estimation by using a single ArUco marker. Distance to the



marker was calculated from the position vector. In the consequent series of measurements, three components of the translation vector were taken under the scope.

III. VERIFICATION TESTS

As the design concept shows the relative position of a UAV is estimated by using three translation vectors. For the verification test each component was verified and corrected according to the following test plan:

- the distance vector value calibration;
- the height vector value calibration;
- the deviation vector value calibration;

A. Distance measurement

The measurement procedure to scale the distance vector was conducted by placing a single ArUco marker in the distance from 15 meters (which is the minimum distance with 25 mm lens) to 50 meters with the 4-meter step between the series of measurements (the last measure point was placed at 50 meters instead of 51 meters). The results of the measurement, presented in figure 4, are proportional to the real distance – measured with the measuring tape. Based on results the correction function was calculated as follows:

$$x_c = 0,54903 * x_m + 1,47615 \quad (3)$$

Where:

x_c – corrected distance to the ArUco marker;

x_m – measured distance to the ArUco marker;

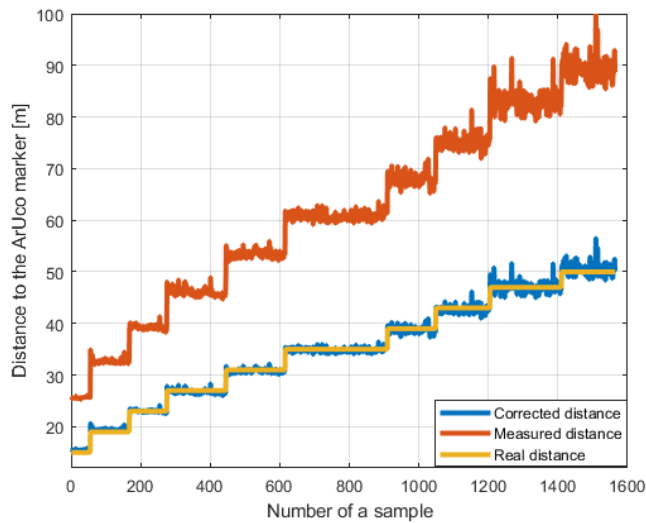


Figure 5. Distance measurements results

As shown in figure 5 the corrected distance value is similar to the real value on the entire detecting range. Although, the

error increase with distance. This occurs due to the low resolution of the camera. The small error of the contour detection occurs in a few meters error – in close distance, single pixels errors do not affect the measurement result so much. The standard deviation of the results is 0,74% for the 15 meters and increases with the distance to 2,09% for the 50 meters.

B. Height measurement

The measure of the height is crucial during the final approach of the UAV. After the calibration, the height vector must be accurate. Before the touchdown information about the UAV vertical distance is crucial to avoid structure and undercarriage damage. The tests have been carried out with three markers placed on the rod with a distance of 1,5 meters between the markers with ID-0 and ID-15 and 1 meter between markers with ID-15 and ID-4 as shown in figure 6.



Figure 6. Height vector measurements procedure

Markers were placed in a distance from 20 to 65 meters with 5 meters step. From the result the correction function was calculated as follows:

$$h_c = h_m - \frac{x_c - 0,5945}{11,114} \quad (4)$$

Where:

h_c – corrected height value over the ArUco marker;

x_c – corrected distance to the ArUco marker;

h_m – measured height value over the ArUco marker;

In figure 7ab, the histogram of the corrected samples is shown. In those figures, it can be noticed that the Gaussian distribution was acquired in most of the data sets.

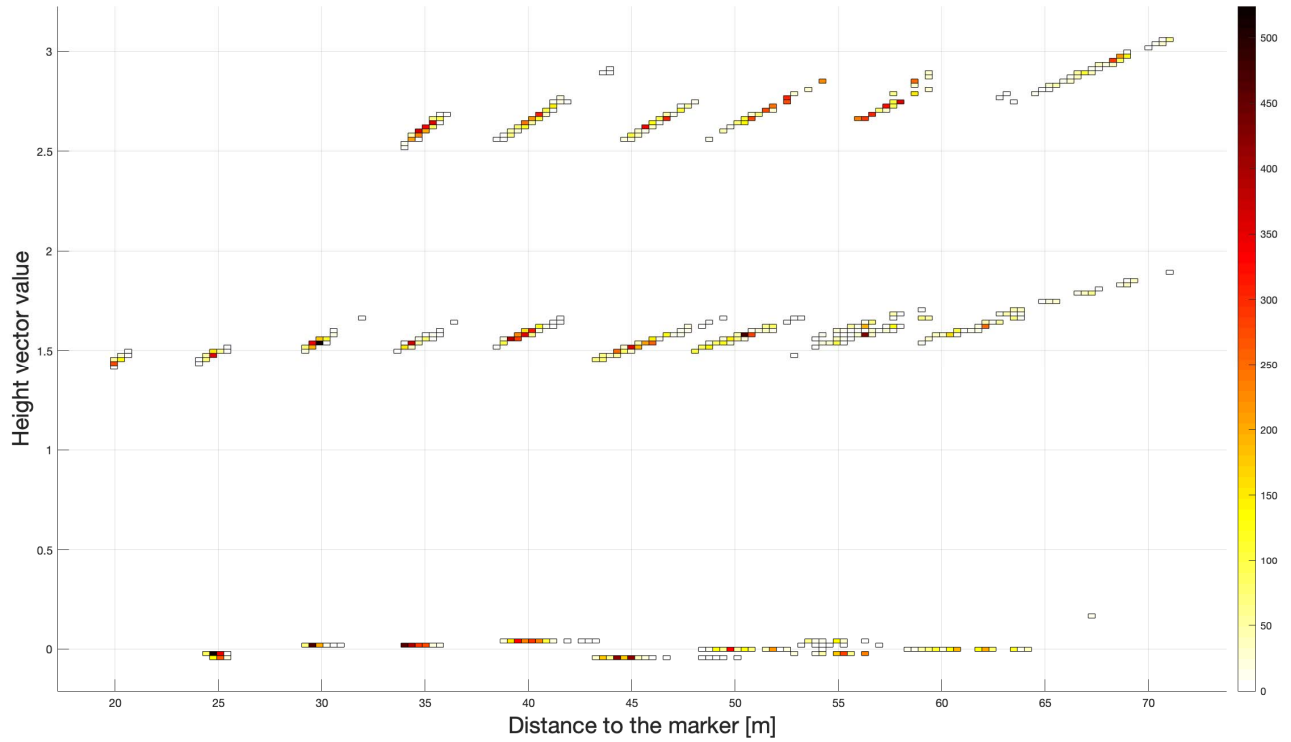


Figure 7. Height vector value in the distance function with the histogram as a colormap

The results in figure 7 show that the height between the measured markers is constant in the entire range. Although, the error increases with the distance up to approximately 10% error at the distance 60 meters from the camera. Figure 8 presents the distribution of samples.

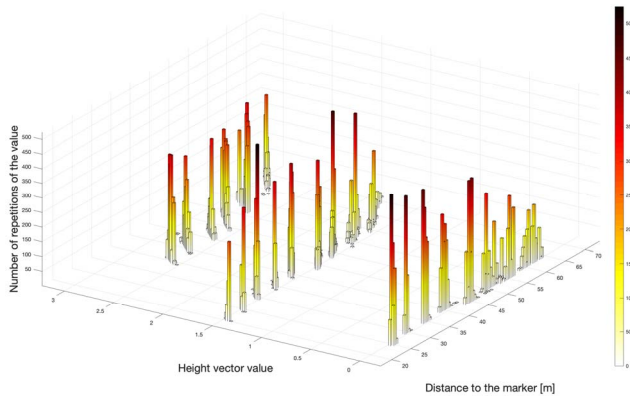


Figure 8. Histogram of the height vector in the distance function

As shown in figure 8 for each data set the Gaussian distribution was acquired. As shown in figure 7 every data set has the value with many repetitions and the area with much fewer repetitions around it. In low-speed relative movement, the error can be filtered by using the distribution of the samples (some consequent values can be averaged). At the end of the range, the value is less consistent. **Reliable pose estimation can be assessed in a range of 15-60 meters.**

C. Deviation measurement

The last component of the position vector is crucial to determine the position of a UAV relatively to the runway axis. According to the design concept presented previously the relative 3D position will be calculated to the multiple markers placed along the runway sides. In the test plan, one marker was placed along the runway axis and four markers were placed along the sides of the road. The first row of the markers was placed on the 45 meters and the next row was placed on the 55 meters.

From the result the correction function was calculated as follows:

$$d_c = d_m - \frac{x_c + 1,4893}{16,537} \quad (4)$$

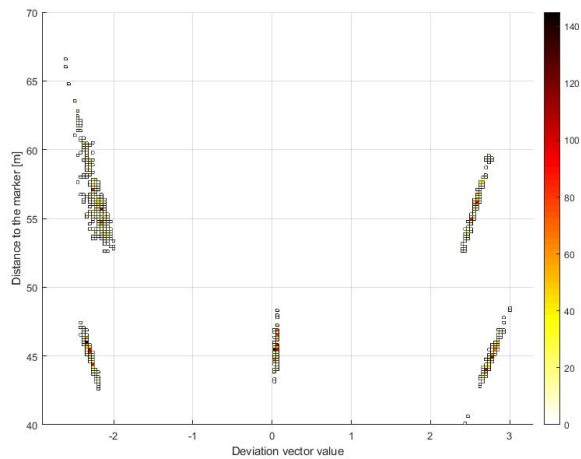
Where:

d_c – corrected deviation value of the ArUco marker to the camera axis;

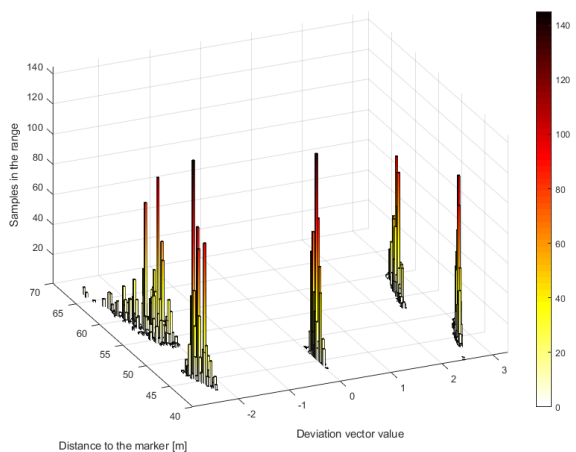
x_c – corrected distance to the ArUco marker;

d_m – measured deviation value of the ArUco marker to the camera axis;

In figure 9, a) the histogram of the samples is presented as 2-dimensional to show the error of the samples and b) 3-dimensional plot to present the distribution of samples.



a)



b)

Figure 9. Histogram of the deviation vector in the distance function: a) 2-dimensional view; b) 3-dimensional view with the distribution of samples

As shown in figure 9 the error of the deviation value is less than 30 cm. For every measurement set, the Gauss distribution was acquired. During the landing, the axis of the runway can be determined by the difference of the deviation vector value calculated from the consecutive rows of the ArUco markers.

IV. CONCLUSIONS

This paper is a presentation of the landing aid system for fixed-wing UAV which operates in BVLOS flights. The navigation algorithm is based on computer vision marker detection and pose estimation. ArUco markers placed along the runway give accurate information about the position of the UAV which is used for the glide slope calculation. Moreover, the proposed infrastructure can be used as a navigation aid during the taxiing on the runway. The test covers the verification of distance measurement accuracy with a single marker and the correction function

calculations. Corrected distance error value was lower than 10% at the range of 60m. The accuracy of the height and deviation from the camera axis was also tested and after implementing the correction functions the 10% error rate was acquired. In the next step, the 3D pose estimation accuracy will be tested along with the dynamic characteristics. Furthermore, the improvement of the pose estimation with an increasing number of visible markers is to be verified. After the implementation of the ArUco landing aid system into the UAV navigation system the pose calculated with ArUco markers can be filtered against a gross error in consecutive samples. This can be achieved by data fusion with the IMU data. Using a short term pose estimation accuracy delivered by IMU the errors of the ArUco pose can be corrected. On the other hand, the IMU error increases in time and can be corrected with the relative pose determined by the ArUco markers. Further development of the ArUco landing aid system covers an upgrade of a camera to the high resolution and more powerful single board computer to provide high frame rate.

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